Recommendation System for E-commerce Platform

Software Project Lab 3 Technical Report

Submitted by
Sakib Al Hasan
BSSE-1209

Institute of Information Technology
University of Dhaka

Supervised by
Dr. Rezvi Shahariar
Associate Professor

Institute of Information Technology
University of Dhaka

Submission Date: 1 December 2024

Letter of Transmittal

1 December, 2024
BSSE 4th Year Exam Committee
Institute of Information Technology
University of Dhaka

Subject: Technical Report Submission - Recommendation System for E-commerce Platform

Sir,
I am submitting the technical report for our Software Project Lab 3, "Recommendation System for E-commerce Platform."
This report comprehensively covers the project's technical details, including design, implementation, and testing phases. While I've tried to do my best, I welcome your feedback for improvement.

Thank you for your consideration.

Sincerely,
Sakib Al Hasan
BSSE-1209
Institute of Information Technology
University of Dhaka

—----------

Supervisor's Signature

Acknowledgement

I am grateful to the Institute of Information Technology for giving me the opportunity to work on the project "Recommendation System for E-commerce Platform". I would like to thank my supervisor, Dr. Rezvi Shahariar, Associate Professor, Institute of Information Technology, University of Dhaka, for his guidance and support throughout the project. His expertise and insights were instrumental in shaping the direction of the project. Lastly, I would like to thank my classmates. They have always been helpful and provided valuable insights from time to time.

Abstract

E-commerce platform popularity increases day by day. In e-commerce platforms the most expected behaviour is recommendation. There are very challenges to apply recommendation in e-commerce systems like limited data of user interaction, customer preference diversity, and new user recommendation.

This e-commerce recommendation platform analyzes user behaviour with user interaction like , dislike, review, purchases etc. It also checks user similarity to recommendation. This also gives real time product suggestions when a user interacts with a product then the user's preference and recommendation will change. For new users this system will give suggestions to popular products that are bought the highest time in this ecommerce site.

It also helps merchants to identify trending products, user preference products and user interests. With this type of recommendation this system will increase the product's sales.

Table of Contents

Recommendation System for E-commerce Platform **0**
Chapter-1: Introduction **5**
1.1 Motivation 5
1.2 Scope 5
1.3 Project Timeline 6
1.4 Purpose of Document 6
Chapter-2: Project Description **7**
2.1. Quality Function Deployment 7
2.1.1. Normal Requirements 7
2.1.2 Expected Requirements 7
2.1.3 Exciting Requirements 7
2.2 Usage Scenario 8
Chapter-3: Scenario Based Modeling **9**
3.1 Use Case Diagram 9
3.1.1 Recommendation system for E-commerce platform 9
3.1.2 Module of E-commerce platform 10
3.1.3 Module of Recommended Product 11
Chapter-4: Data Based Modeling **13**
Chapter-5: Class-Based Modeling **17**
5.1 Analysis Classes 17
5.2 Class cards 18
Chapter-6: Architectural Design **22**
6.1 Architectural Context Diagram 22
6.2 Defining Archetypes 22
Model View Controller (MVC) Architecture Pattern 23
6.3 Top Level Components 23
Chapter-7: Preliminary Test Plan **25**
7.1 High Level Description of Testing Goals 25
7.2 Test Cases 25
Chapter-8: Conclusion **28**
Chapter-9: References: **29**

Chapter-1: Introduction

In the realm of modern e-commerce, recommendation systems have become an integral part of enhancing user experience and driving business growth. These systems leverage advanced algorithms to analyze user behavior and preferences, providing personalized product suggestions that resonate with individual tastes. The importance of such systems cannot be overstated, as they significantly influence customer satisfaction, engagement, and sales conversion rates.

The Recommendation System for E-commerce Platform steps in to bridge the gap between user needs and business goals. By analyzing user interactions, preferences, and contextual factors in real time, the system delivers dynamic and relevant product suggestions. Whether it's helping users discover new products, improving their browsing experience, or enabling merchants to optimize offerings, this system plays a pivotal role in transforming online shopping into a more personalized and intuitive journey.

1.1 Motivation

The motivation behind developing this recommendation system arises from the growing demand for personalized shopping experiences in the e-commerce domain. With millions of products available across diverse platforms, users often face difficulty navigating and finding items that match their interests. Simultaneously, e-commerce businesses are seeking innovative ways to enhance customer engagement, improve conversion rates, and gain a competitive edge.

Modern consumers expect platforms to anticipate their needs and preferences. However, challenges such as handling sparse data for new users or products (the cold-start problem) and ensuring recommendations are both accurate and diverse persist. The Recommendation System for E-commerce Platform addresses these challenges by utilizing cutting-edge machine learning techniques, collaborative filtering, and contextual data analysis to provide tailored suggestions.

Furthermore, the system takes inspiration from research in recommendation algorithms and real-world applications to ensure high efficiency and relevance. By continuously refining the recommendation models based on user feedback and behavioral data, the system aims to create a seamless and enjoyable shopping experience for users while boosting business outcomes for merchants.

1.2 Scope

The scope of recommendation is given below:

- It will be an ecommerce website
- Gives recommendation to user according their behaviour

1.3 Project Timeline

The estimated timeline of recommendation system is given in figure-1 below:



Figure-1: Timeline of Recommendation System

1.4 Purpose of Document

The purposes of this document are-

- Identify and analyze the requirements
- Design the architecture
- Design the test plan
- Describe methodology
- Reduce the development effort
- Improve understanding

Chapter-2: Project Description

The following outlines the overview of the Quality Function Deployment (QFD) for the **Recommendation System for E-commerce Platform**. The journey toward defining the QFD began with an in-depth review of relevant research literature, including state-of-the-art methodologies for recommendation systems and their applications in e-commerce. These reviews provided valuable insights into current trends and best practices in the domain. This was further enriched by proactive stakeholder engagement, involving key players in the e-commerce industry, data scientists, and software developers. Their perspectives and concerns were thoroughly analyzed to ensure the system addresses real-world challenges and aligns with user expectations. Stakeholders from diverse backgrounds, including online retailers and technical experts, played a crucial role in shaping the vision and goals of the project.

2.1. Quality Function Deployment

Quality Function Deployment (QFD) is a structured method used in software requirement analysis to transform user requirements into technical specifications.
Requirements are identified for this ecommerce platform-

2.1.1. Normal Requirements

1. **Add product in cart:** Customises can add product in cart that they are interested in buying.
2. **Order a product:** From cart users can buy after ordering a product.
3. **User feedback collection:** Use feedback to refine recommendations through a feedback loop like- rating, like/dislike etc.
4. **Cross-platform compatibility:** The system should be responsible on web and mobile platforms.
5. **All user support:** Both login users and anonymous/guest users can browse this platform.
6. **Recommendation:** Users get product recommendation in home age and through notification.

2.1.2 Expected Requirements

1. **Cold Start Solution:** It should recommend new users or products with little interaction history.
2. **Insights for merchants:** It should provide an analytics dashboard for merchants, like- trending product, popular user preference and purchase pattern. Also that can update product details.
3. **Dynamic real-time recommendations:** Event-driven architecture; update recommendations as users browse products dynamically.

2.1.3 Exciting Requirements

1. **User Behavioural Analysis:** Customises product recommendations for each user according to their preferences and behavior.

2. **Diversity in recommendations:** Diversify algorithms to avoid repetitive suggestions such that users are not bored.

2.2 Usage Scenario

The **Recommendation System for E-commerce Platform** will enhance the shopping experience by providing personalized, real-time product suggestions to users while offering merchants valuable insights. Usage scenarios for the system are segmented as follows:

Installation and Integration: E-commerce platforms can integrate the recommendation system into their existing infrastructure. The system will seamlessly connect with standard platform features like search, filters, ratings, and analytics dashboards.

Personalized Product Suggestions: The system will analyze user preferences, browsing history, and purchase patterns to recommend relevant products. For logged-in users, recommendations will be tailored based on their past interactions, while guest users will receive trending or popular product suggestions.

Real-Time Updates: As users browse the platform, the recommendation system will dynamically update suggestions in real time. For instance, when a user views or adds a product to the cart, the system will immediately suggest complementary or related items.

Cold-Start Problem Handling: For new users or products with minimal interaction history, the system will provide recommendations based on trending products, category popularity, or demographic data, ensuring a seamless experience from the start.[1]

Merchant Insights: Merchants will receive analytics on user behavior, such as trending products, most-viewed categories, and customer preferences. These insights will help merchants optimize their offerings and marketing strategies.

Feedback Mechanism: Users can provide feedback by liking, disliking, or rating products. This feedback will be used to refine recommendations and improve system accuracy over time. Merchants can also share insights on how recommendations impact their sales and product visibility.

Notification of New Recommendations: The system will notify users about newly recommended products via personalized banners, notifications, or emails, ensuring continuous engagement and discovery of new items.

Chapter-3: Scenario Based Modeling

Scenario-based modeling is an approach used in software development and design to create models that depict how a system behaves or reacts in various real-world situations or scenarios. These scenarios are typically based on user interactions, events, or conditions and help developers understand and visualize how the software will function in different contexts. By simulating these scenarios, developers can better identify potential issues, make informed design decisions, and ensure that the software meets user requirements across different usage contexts.

3.1 Use Case Diagram

A use case diagram is a visual representation in software engineering that illustrates how a system interacts with its external entities or actors to achieve specific goals or tasks. It shows the relationships between use cases (representing system functionalities) and actors (representing external users or systems). Use case diagrams are typically created through requirements gathering, where system functionalities and user interactions are identified, and then diagrammed using specialized software modeling tools.

3.1.1 Recommendation system for E-commerce platform

Here figure-2 describe the whole recommendation system for e-commerce platform with actors.

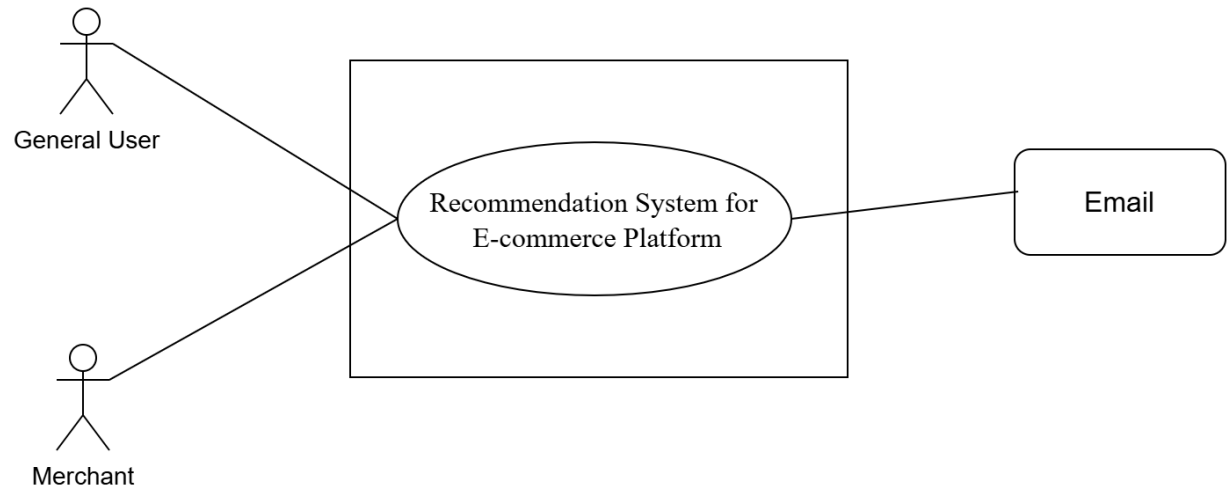


Figure-2: UseCase Level 0

Primary actors: General User, Merchant
Secondary actors: Email
Goal in Context: The diagram shown above represents the whole Recommendation system of E-commerce platform.

3.1.2 Module of E-commerce platform

Figure-3 breaks the usecase level 0 and explains further modules how to interact with actors.

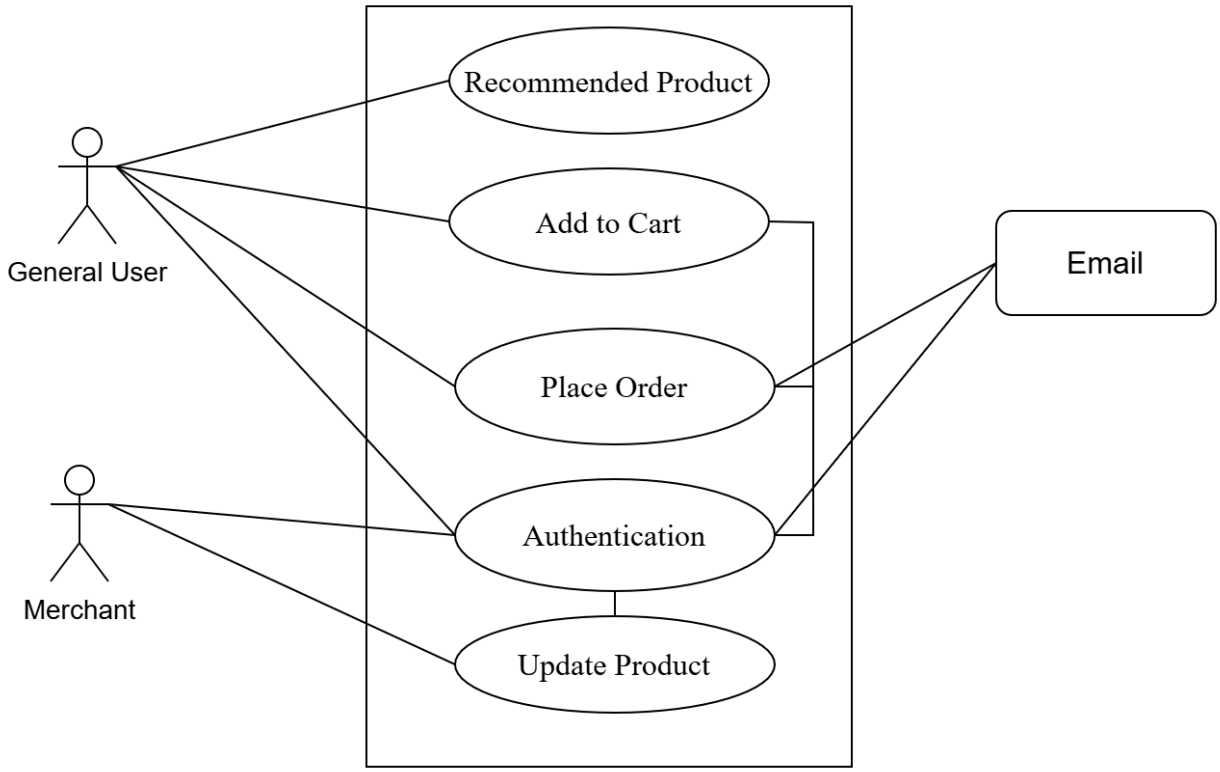


Figure-3: UseCase Level 1

Primary Actor: General User, Merchant
Secondary actors: Email
Goal in Context: The diagram shows the modules of the Recommendation system. It consists of 5 modules.
The modules in figure-3 can be elaborated as follows-

3.1.2.1 Recommendation Product: It is an entry point for users. When users visit this e-commerce site then they find products in home pages. This product is a recommended product and it will be generated using user behavioural analysis. It will analyze user behaviour to generate dynamic and valuable recommendation products.

3.1.2.2 Add to cart: After browsing product home pages, users can add it to the card. Here the cart is a preferable place that the user may be interested in this product and may be able to buy it. To add products in cart, users must login to this system.

3.1.2.3 Place Order: After adding to cart, users can buy it. Users can pay this user using a payment gateway and add an address to ship the product.

3.1.2.4 Authentication: For better recommendation and place his/her order and add to cart feature user must be login to this system.

3.1.2.5 Update Product: Regular product update and monetization merchant must update ecommerce site product.

3.1.3 Module of Recommended Product

Here we break usecase level-1 recommended product module and describe in figure-4 how it works and interact with users.

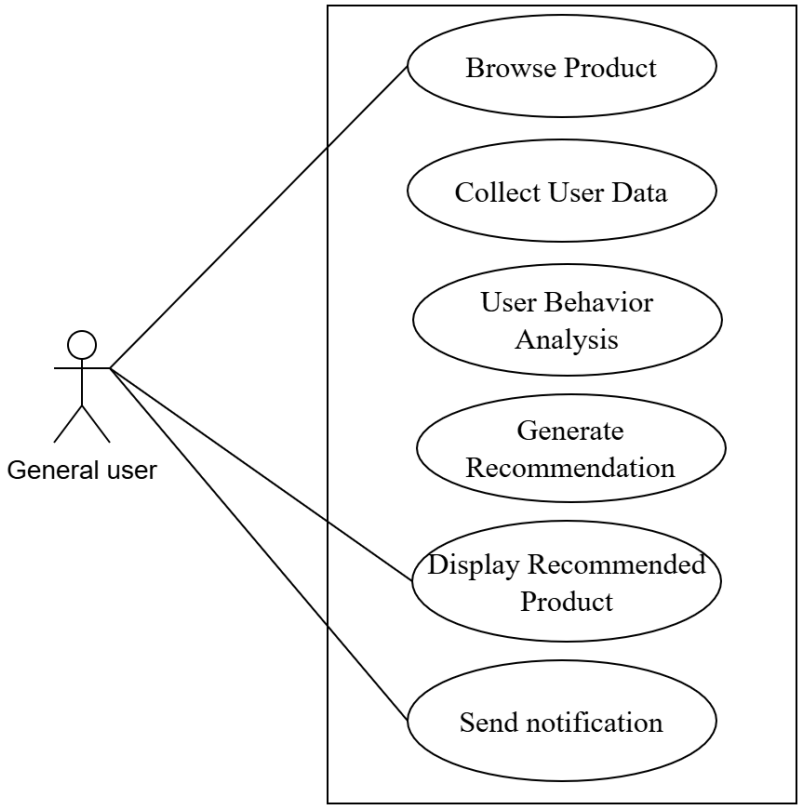


Figure-4: UseCase Level 2

Primary Actor: General User
Goal in Context: The diagram shows the modules of the Recommendation system. It consists of 5 modules.
The modules can be elaborated as follows

3.1.3.1 Browse Product: Users will find products using this site. User behaviour will collect through user product browsing.

3.1.3.2 Collect User data: User data is mainly stored in the database and some data will be stored in the browser for guest users.

3.1.3.3 User Behaviour Analysis: After collecting data, it will preprocess data and analyze user behaviour.

3.1.3.4 Generate Recommendation: According user behaviour analysis it will generate most user recommendation products using different algorithms.

3.1.3.5 Display Recommendation Product: Lastly it shows a product recommendation and the user will get this product through browsing the site.

3.1.3.6 Send Notification: If a product becomes more relevant to the user then it will send it to user as notification.

Chapter-4: Data Based Modeling

Database modeling is a visual representation of a database structure and how data is organized within it. It involves creating entity-relationship diagrams (ERDs) that depict tables (entities), attributes (columns), and the relationships between them. Database modeling is typically achieved by analyzing the data requirements of a system, identifying entities and their attributes, and then using modeling tools or software to create ERDs that serve as a blueprint for database design and implementation.

The entities of Recommendation System after analyzing the usage story are:

- User
- Product
- Category
- Brand
- Cart
- Feedback
- Search
- View
- Notification

Here is the E-R diagram in figure-5 that describes how entities are connected to each other with their relation. In each entity there is a primary key and foreign key is given. There are also weak relations that are shown in the double rectangle and double diamond box. In each relation there is a related relation name.

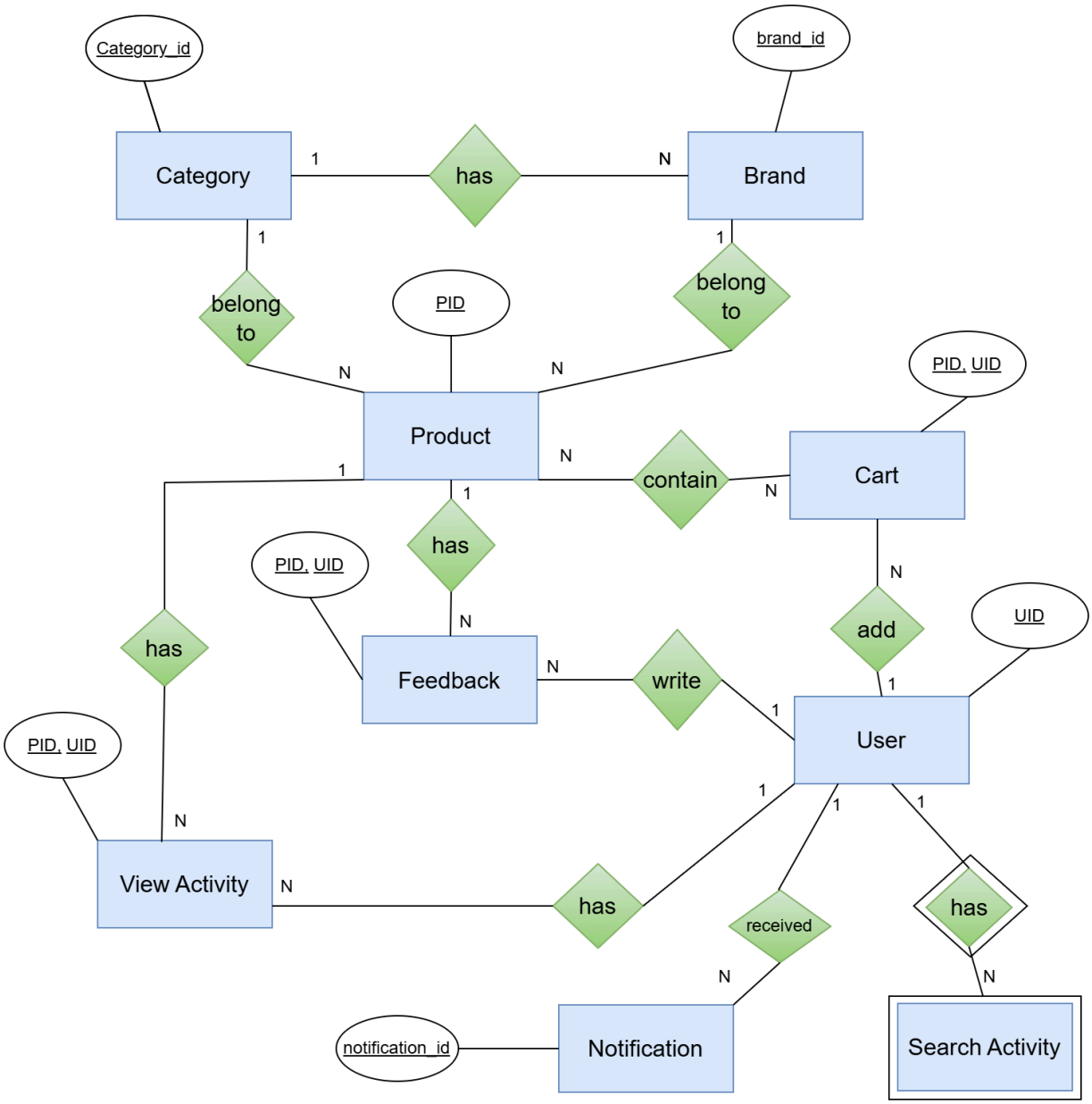


Figure-5: ER-Diagram

The schemas derived from the ER-diagram is given in table-1:

Table-1 : Schema of Recommendation System

Schema	Attributes	Type
User	UID	varchar(30)
	Email	varchar(50)
	Password	varchar(50)

	create_time	date-time
	user_type	varchar(1)
Product	PID	varchar(30)
	description	varchar(255)
	model	varchar(50)
	price	number
	stocks_item	number
	BID	varchar(30)
	category_id	varchar(30)
	avg_rating	number
	item_purchase	number
	item_view	number
	cluster	number
Category	category_id	varchar(30)
	category_name	varchar(50)
Brand	BID	varchar(30)
	brand_name	varchar(50)
	category_id	varchar(30)
Feedback	PID	varchar(30)
	UID	varchar(30)
	create_time	date-time
	rating	number
	comment	varchar(255)
Search	UID	varchar(30)
	key_word	varchar(255)
View	PID	varchar(30)
	UID	varchar(30)

	time_visit	number
	last_visit	date-time
Cart	PID	varchar(30)
	UID	varchar(30)
	quantity	number
	time	date-time
	status	varchar(1)
Notification	notification_id	varchar(30)
	UID	varchar(30)
	content	varchar(255)
	status	boolean
	create_at	date-time

Chapter-5: Class-Based Modeling

Class-based modeling is a visual representation in software engineering that illustrates the structure of a software system using classes, their attributes, and relationships. It is commonly used in object-oriented design and modeling.

5.1 Analysis Classes

After identifying nouns from the scenario, I have filtered nouns belonging to the solution domain using General Classification (External entities, Things, Events, Roles, Organizational units, Places, and Structures). Nouns selected as a potential class were filtered using Selection Criteria (Retained information, Needed services, Multiple attributes, Common attributes, Common operations, and Essential requirements). After performing analysis on potential classes, I have found the following analysis classes

- RecommendationSystem
- User
- GuestUser
- Product
- Merchant
- Feedback
- AnalyticsDashboard
- Category
- Algorithm
- Notification
- Cart
- Interaction
- Scenario
- Event
- Notification

Science RecommendationSystem and Algorithm have the same responsibility so we can reduce class. There is also the same responsibility that is Interaction, Scenario and Event class. Here guest user and Marchant are inherited from User class. So our final classes are-

- User
- GuestUser
- Product
- Merchant
- Feedback
- AnalyticsDashboard
- RecommendationSystem
- Cart
- Notification

5.2 Class cards

The class cards of Recommendation System is given below:

Table-2: CRC card(User)

User	
Attributes	**Methods**
-Id -name -email -password -createTime -userType	+register() +login() +forgotPassword() +changePassword +updateInformation +viewDashboard() +orderProduct()
Responsibilities	**Collaborators**
● Users can view product details and order products from cart after adding them. ● Also give feedback to the product.	Product Cart Feedback

Table-3: CRC card(Guest User)

Guest User	
Attributes	**Methods**
-browsingHistory	+setDataAfterLogin()
Responsibilities	**Collaborators**
● Guest user inherit User	User

Table-4: CRC card(Merchant)

Merchant	
Attributes	**Methods**
-List<Product>	+updateProduct() +addProduct() +deleteProduct()
Responsibilities	**Collaborators**
● Merchant inherit to user	User

• Merchant can update,add,delete product • Merchant has analytics dashboard	Product AnalyticsDashboard

Table-5: CRC card(Product)

Product	
Attributes	**Methods**
-product_id -description -model -price -stocks_item -brand_id -category_id -avg_rating -item_purchase -item_view -cluster	+addToCart() +reviewProduct() +storeUserInteraction()
Responsibilities	**Collaborators**
• Products are added to cart • Products are used in recommendation for suggest to user	Cart Recommendation

Table-6: CRC card(Feedback)

Feedback	
Attributes	**Methods**
-product_id -user_id -rating -comment	+saveFeedback()
Responsibilities	**Collaborators**
• Use can write feedback on product	User

Table-7: CRC card(RecommendationSystem)

RecommendationSystem	
Attributes	**Methods**

-purchaseProduct -visitProduct -cartAddedProduct -likedProduct	+createPreferenceMatrix() +createSimilarProduct()
Responsibilities	**Collaborators**
• Suggest recommendation product to user	User

Table-8: CRC card(AnalyticsDashboard)

AnalyticsDashboard	
Attributes	**Methods**
-merchant -trendingProducts -mostViewedCategories -customerPreferences	+generateInsights() +updateTrendingProducts() +trackCustomerBehavior()
Responsibilities	**Collaborators**
• It shows trending and user preference products.	RecommendationSystem

Table-9: CRC card(Cart)

Cart	
Attributes	**Methods**
-addedProduct -orderedProduct -canclledProduct -userId	+addToCard() +orderedProduct() +removeFromCard()
Responsibilities	**Collaborators**
• Add product into cart • Users order products from cart.	Product User

Table-10: CRC card(Notification)

Notification	
Attributes	**Methods**

-userId -message -time -status	+generateNotification() +sendNotification()
Responsibilities	**Collaborators**
• Notification will send to user	User

Here is in figure-6 the CRC diagram that represents how classes are connected to each other and also shows connection with their appropriate relation name.

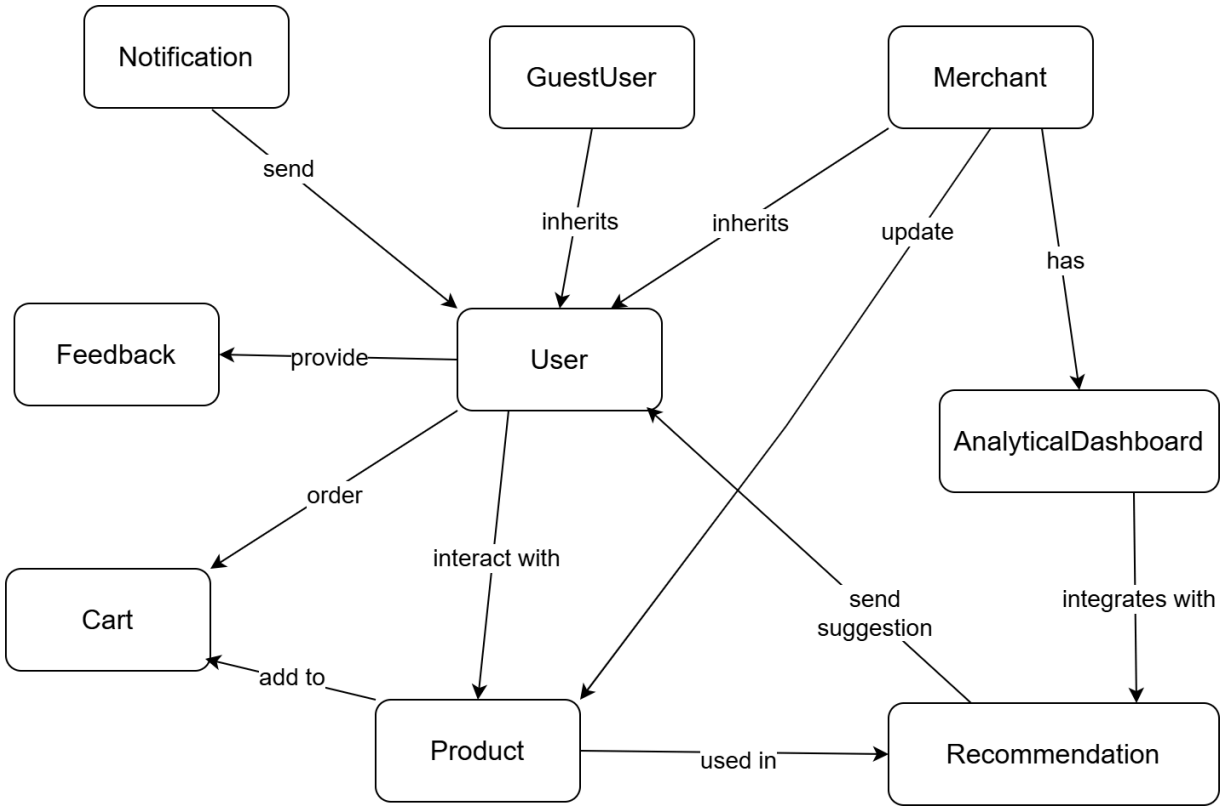


Figure-6 : CRC diagram

Chapter-6: Architectural Design

Architectural design is a visual representation in software engineering that outlines the high-level structure and organization of a software system. It focuses on defining the major components or modules of the system, their interactions, and the overall system's architecture.

6.1 Architectural Context Diagram

The architectural context diagram for the tool is given below in figure-7. That shows that how user interact with the system and system depends on the user.

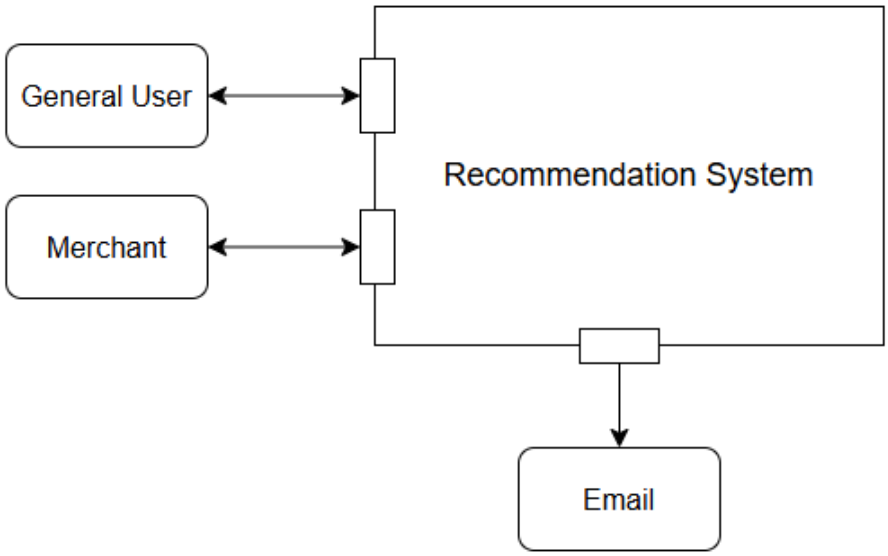


Figure-7: Architectural Context Diagram

Two categories of users are supposed to be the target for the tool. They are software general users and merchants. Here merchants are producers and general users are consumers of the e-commerce site. There is one subordinate system, Email. Email will be sent to the user for password change and further information.

6.2 Defining Archetypes

We have derived our archetypes by examining the analysis classes. We have found the following archetypes:

Model View Controller (MVC) Architecture Pattern

Here is figure-7 that describes a pattern that I will implement in my project. This pattern is mvc pattern. This is a general view of MVC patterns.

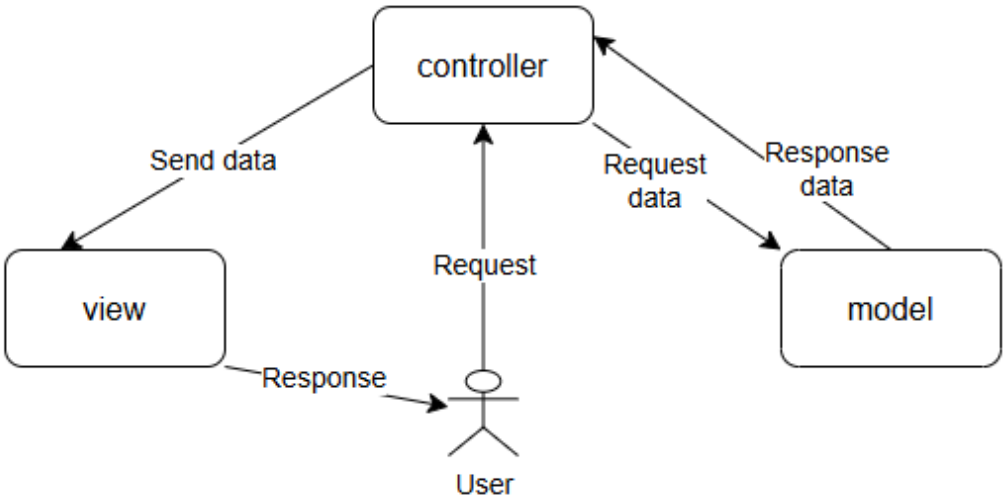


Figure-7: MVC architecture pattern

View: Responsible for presenting data to users and handling user interactions. Display recommendations, analytics, and product details dynamically. Capture user feedback (visit, ratings) and interactions.

Controller: Acts as a bridge between the Model and View, processing user requests and updating the view. Process user actions like product views, feedback submission, and purchases. Call the appropriate model logic and update the UI dynamically.

Model: Responsible for managing the data, logic, and rules of the application. Store and manage user preferences, product details, feedback, and analytics. Handle data-related operations like CRUD, business rules, and database interactions.

6.3 Top Level Components

Overall architectural structure with top level components is illustrated below in figure-8

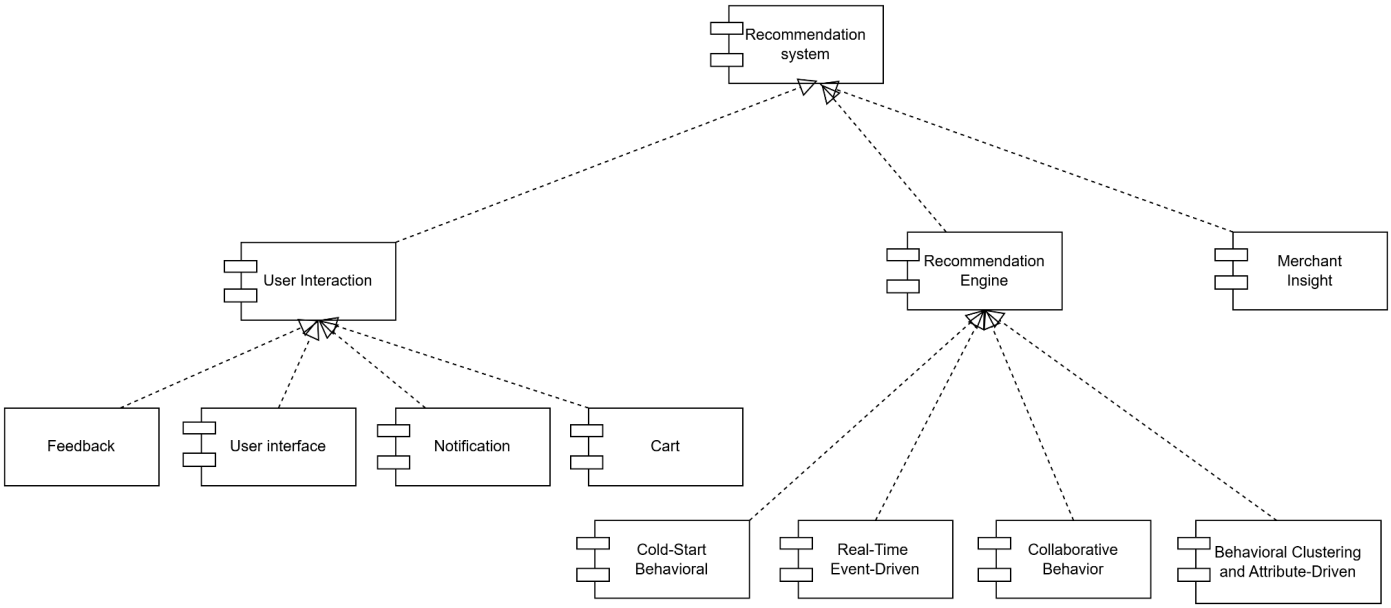


Figure- 8: Top Level Components

Chapter-7: Preliminary Test Plan

In this chapter, a high level description of testing goals and summary of features to be tested are presented.

7.1 High Level Description of Testing Goals

Testing goals for this tool is represented below from a high level-

- To demonstrate that the tool meets its requirements
- To ensure that it produces correct response most of the time
- To ensure the features of the project are extracted correctly
- To ensure that the tool runs smoothly

7.2 Test Cases

Test cases for the testing of this tool are documented below-

- **Test Case 1:** Behavioral Clustering

  Test Scenario:
  Simulate user interactions with multiple products from the same category.
  Run the clustering algorithm to assign users to behavioral clusters.

  Expected Outcome:
  Users are correctly grouped into clusters with similar behavioral patterns.

- **Test Case 2:** Update cart item products

  Test Scenario:
  Add and remove product in cart item.
  Order product from cart item.

  Expected Outcome:
  Products will correctly add, remove or order in the cart.

- **Test Case 3:** Attribute-Driven Recommendations

  Test Scenario:
  User interacts with products having specific attributes (e.g., price <$50).
  Generate recommendations based on attribute filtering.

Expected Outcome:

Recommendations align with the user's preferred product attributes.

- **Test Case 4:** Cold-Start Recommendations for New User

  Test Scenario:

  Create a new user account and browse the platform without interacting with any product.
  Allow the system to generate recommendations.

  Expected Outcome:

  Recommendations are based on trending or popular products.

- **Test Case 5:** Cold-Start Recommendations for New Product

  Test Scenario:

  Add a new product to the inventory with minimal metadata.
  Check its visibility in the recommendations for a random user.

  Expected Outcome:

  The new product is recommended based on its category or initial metadata.

- **Test Case 6:** Real-Time Event-Driven Updates

  Test Scenario:

  User views a specific product.
  Check the dynamically updated recommendations list in real time.

  Expected Outcome:

  Recommendations are adjusted dynamically to include relevant products.

- **Test Case 7:** Diversity in Recommendations

  Test Scenario:

  User interacts repeatedly with products from the same category.
  Check the variety of products in the recommendations list.

  Expected Outcome:

  Recommendations include a diverse range of products across different categories.

- **Test Case 8:** Feedback-Driven Refinement

  Test Scenario:

  User provides feedback (review a product).
  Generate recommendations after considering the feedback.

Expected Outcome:

The disliked product is deprioritized in future recommendations.

- **Test Case 9:** API Gateway Responses

  Test Scenario:

  Send valid API requests for product recommendations.
  Check the API response structure and data.

  Expected Outcome:

  API returns correct recommendations in the expected format.

- **Test Case 10:** Analytics Dashboard for Merchants

  Test Scenario:

  Log in as a merchant and access the analytics dashboard.
  Verify displayed insights (trending products and user behavior metrics).

  Expected Outcome:

  The dashboard displays accurate, up-to-date analytics relevant to the merchant.

- **Test Case 11:** Search and Recommendations Integration

  Test Scenario:

  Search for a product and check if relevant recommendations appear based on the search query.

  Expected Outcome:

  Search results display, and related products are shown in the recommendation section.

- **Test Case 13:** Notification System

  Test Scenario:

  Check if users receive notifications or banners for newly recommended products.

  Expected Outcome:

  Notifications appear for logged-in users based on preferences or browsing history.

Chapter-8: Conclusion

In this report, I have discussed all the necessary technical details to develop the “E-commerce Recommendation System” aimed at enhancing user experience through personalized and data-driven product suggestions. I have elaborated on the scope, assumptions, and usage scenarios of the system. Scenario-based modeling, data modeling, and class-based modeling have been implemented to ensure robust system design. The architectural design has been presented in detail, accompanied by high-level descriptions of the testing goals and summaries of the features and items to be tested. Figures and diagrams have been included throughout the report to facilitate a comprehensive understanding of the system's technical aspects.

Although the current recommendation system provides dynamic, personalized, and behavior-driven recommendations, there is room for further improvement. Potential enhancements could include the integration of advanced machine learning algorithms, real-time user behavior prediction, and the support for cross-platform recommendations. These upgrades can ensure the system remains scalable, adaptive, and competitive in rapidly evolving e-commerce landscapes.

With this document, I have aimed to provide a clear and concise understanding of the workflow, architecture, and functionality of the system. I hope this report equips the reader with the necessary insights into the technical implementation and potential future developments of the recommendation system.

Chapter-9: References:

[1] A survey of explainable e-commerce recommender system. In 2022 IEEE/WIC/ACM International Joint Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT)
[2] AnE-Commerce Recommendation System Based on Dynamic Analysis of Customer Behavior. https://www.mdpi.com/2071-1050/13/19/10786